

Summary Recommendations

Customer Churn Analysis – Detailed Summary

This project focuses on analyzing customer churn behavior in a telecom company using Python and data analysis libraries such as Pandas, Matplotlib, and Seaborn. The purpose of the project is to identify the major reasons behind customer churn and discover patterns that affect customer retention.

Project Objective

The main goal of this analysis is to:

- Understand customer behavior
 - Identify factors responsible for churn
 - Compare churn across different customer groups
 - Help telecom companies improve customer retention strategies
-

Data Cleaning & Preprocessing

Several preprocessing steps were performed before analysis:

- Checked dataset structure and column information
- Removed or handled missing values in **TotalCharges**
- Converted data types where required
- Checked duplicate records
- Converted categorical values into readable format
- Verified null values and data consistency

These preprocessing steps improved data quality and made visualization more accurate.

Exploratory Data Analysis (EDA)

1. Churn Distribution Analysis

A countplot and pie chart were used to analyze overall churn distribution.

Key Insight

- Around **73% customers stayed** with the company.
- Approximately **27% customers churned**.

This indicates that although most customers are retained, the churn percentage is still significant and requires business attention.

2. Gender-wise Churn Analysis

The churn distribution was analyzed across male and female customers.

Observation

- Male and female customers showed nearly similar churn behavior.
- Gender did not strongly impact customer churn.

This suggests that churn depends more on services and contract-related factors rather than gender.

3. Senior Citizen Churn Analysis

A stacked percentage bar chart was created for **SeniorCitizen** vs **Churn**.

Key Findings

- Senior citizens had a noticeably higher churn percentage compared to non-senior customers.
- Percentage visualization made comparison easier.

Business Insight

Senior citizens may require:

- Better customer support
- Simpler plans
- Personalized assistance

to improve retention.

4. Tenure Analysis

Tenure represents the number of months a customer stayed with the company.

Histogram and boxplot visualizations were used to study tenure distribution.

Key Findings

- Customers with low tenure showed higher churn rates.
- Long-term customers were more loyal and less likely to leave.

Business Insight

New customers are more vulnerable to churn during the early months of service. Telecom companies should focus on:

- Early customer engagement
 - Welcome offers
 - Better onboarding experience
-

5. Contract Type Analysis

Customer churn was compared across different contract types.

Major Observation

- Customers with **Month-to-Month contracts** had the highest churn percentage.
- Customers with **One-Year** and **Two-Year contracts** showed much lower churn rates.

Business Insight

Long-term contracts increase customer retention and loyalty.

6. Payment Method Analysis

Different payment methods were analyzed against churn behavior.

Findings

- Customers using **Electronic Check** showed higher churn percentages.
- Automatic payment users had comparatively lower churn rates.

Business Insight

Encouraging automatic payment systems may improve customer retention.

7. Service-wise Churn Analysis

Multiple subplot countplots were created for telecom services:

- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies

Key Insights

- Customers without:
 - Online Security
 - Tech Support
 - Device Protection

- showed higher churn percentages.
- Fiber optic internet users displayed relatively higher churn behavior.

Business Insight

Additional support and security-related services help improve customer loyalty.

8. Internet Service Analysis

Internet service type was one of the strongest churn indicators.

Findings

- Fiber optic users had higher churn rates.
- Customers without internet service had lower churn percentages.

This may indicate:

- Pricing issues
 - Service quality concerns
 - Customer dissatisfaction in fiber plans
-

Percentage-Based Visualizations

Throughout the analysis, percentage-based stacked bar charts were used instead of only raw counts. This improved data interpretation by:

- Showing proportional comparisons
 - Making churn patterns easier to understand
 - Helping identify high-risk customer groups more clearly
-

Overall Business Recommendations

Based on the analysis, the telecom company should:

- Focus on retaining new customers during early tenure
 - Promote long-term contracts
 - Improve customer support for senior citizens
 - Encourage automatic payment methods
 - Improve fiber optic service quality
 - Offer Online Security and Tech Support bundles
 - Provide personalized retention offers for high-risk customers
-

Conclusion

This project successfully analyzed customer churn using data visualization and exploratory data analysis techniques. The combination of countplots, stacked percentage charts, histograms, and service-based comparisons helped uncover meaningful business insights.

The analysis revealed that:

- Contract type
- Tenure
- Internet services
- Security services
- Payment methods

are major factors influencing customer churn.

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